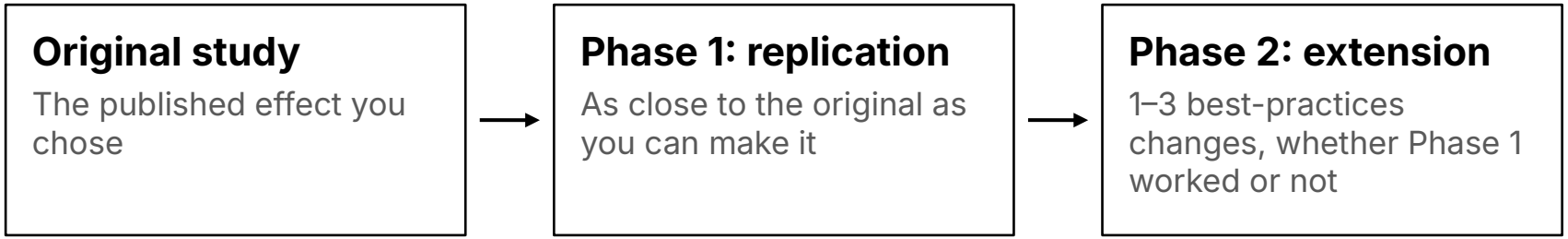
Do best-practices extensions improve replications?

Psych 251 class metascience project, 2026

Proposal and choice points · Monday, November 2

The proposal



The question

Do changes drawn from Experimentology (measurement, design, sample, checks) make a replication more successful, and recover more of the original effect?

Who's in

- Opted-in projects: extra budget, and every opted-in student is an author
- Track 1 only; scientific extensions (Track 2) are excluded
- We expect about 15–20 projects

What we will measure

Replication success

Course staff rate each phase against the original, from 0 to 1. A score of .75 or more counts as a success.

Effect size ratio

Your effect divided by the original effect, both as standardized mean differences. We compute it by re-running your analysis on the data in your repository.

How we compare

Phase 2 against Phase 1, pooled across every project. Boyce et al. (2024) found that 5 of 17 rescues of failed class replications (29%) mostly or fully replicated; we will set our results beside theirs, but the studies differ too much for a formal comparison.

Also: precision (how much the standard error shrank, beyond what sample size alone explains).

Choice points

Staff ratings of replication success

All four of us rate every project: during presentations, or from your report if the data were not in yet.

Rescue or upgrade: who decides?

Is there really a difference?

Ambiguity ratings

How systematic should they be, and what system do we use?

Agent reproducibility audits

When do they happen, and who runs and checks them?

Attitudes, chat logs, comprehension quiz

These study you rather than your experiments, so they require a new IRB protocol.

Plan registered on OSF by Nov 13 · Phase 2 data collected the week of Nov 16